



A DATA-DRIVEN ANALYSIS OF UNEMPLOYMENT AND ITS DETERMINANTS IN JORDAN

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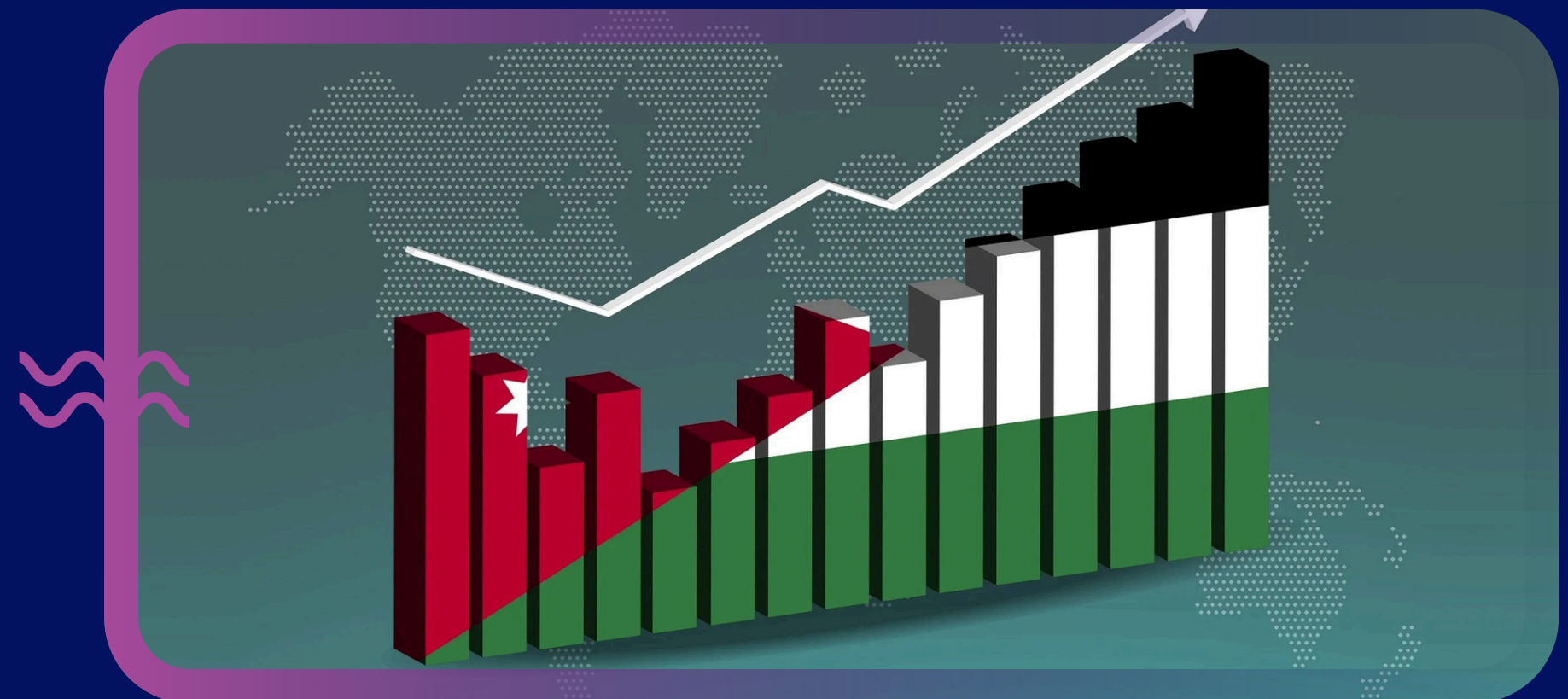


JORDAN'S ECONOMIC LANDSCAPE AND LABOR MARKET CHALLENGES

- However, the Jordanian economy has faced several challenges over the past decade, including population growth, regional instability, pressure on public resources, and limited job creation opportunities.
- One of the most important challenges is unemployment. High unemployment affects economic growth, household income, social stability, and future opportunities, especially among youth and graduates.
- Because unemployment differs across governorates and changes over time, data-driven analysis is needed to understand the problem more clearly and support better decision-making.

This project uses data to understand unemployment patterns and the socioeconomic factors behind them.

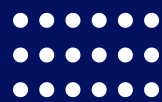
Jordan is a developing economy located in the heart of the Middle East. Despite its limited natural resources, Jordan has built a strong foundation in education, services, tourism, trade, and human capital.





WHY DOES UNEMPLOYMENT MATTER?





PROJECT IDEA

01

Understanding unemployment trends

Analyze unemployment patterns across Jordanian governorates from 2014 to 2024, and understand how unemployment rates changed over time.

02

Identifying key determinants

Study the relationship between unemployment and socioeconomic indicators such as population density, economic establishments, and education-related factors.

03

Applying data-driven models

Use panel data analysis, Pooled OLS, Fixed Effects, VIF, PCA, and clustering techniques to generate reliable analytical insights.

04

Building BI dashboards

Present the results through interactive Tableau dashboards that support visualization, governorate comparison, and data-driven decision-making.





Main Dataset Variables

Variable	Meaning
Unemployment Rate	Main target variable
Population Density	Labor market pressure indicator
Establishments per 1,000	Economic activity indicator
Schools per 1,000 Population	Education infrastructure indicator
Students per Class	Education capacity indicator
Student Ratio 5–19	Education participation indicator

Data Sources

- Jordan Department of Statistics: unemployment, employment, labor force, population.
- Social Security Corporation: number of establishments.
- Ministry of Education: schools, students, classes, education indicators.



Dataset Features

Data Cleaning and Preparation:

- Renaming columns
- Converting data types
- Handling missing values
- Selecting relevant features
- Creating Year Squared variable
- Standardizing variables for clustering
- Exporting cleaned output



Panel Data Structure

Panel Data = Cross-sectional data + Time-series data

In this project, the dataset combines:

1. Cross-sectional dimension

Jordanian governorates

Example: Amman, Irbid, Zarqa, Aqaba, Ma'an, Mafraq...

2. Time dimension

Years from 2014 to 2024

Each row represents:

One governorate observed in one specific year



Why Panel Data?

- Tracks unemployment changes over time.
- Compares unemployment across governorates.
- Supports Fixed Effects modeling.
- Captures regional differences more accurately.



Modeling Techniques

The project used regression models, validation methods, and clustering techniques to analyze unemployment patterns and socioeconomic determinants.

Pooled OLS Model

Pooled OLS was used as the baseline regression model

$R^2 = 0.281$
RMSE = 3.586
MAE = 2.955

The model captured general patterns but had limited explanatory power.

Fixed Effects Model

Fixed Effects was used because the dataset is panel data and governorates have different characteristics.

$R^2 = 0.847$
RMSE = 1.657
MAE = 1.306

Fixed Effects achieved the best performance by controlling for governorate-specific differences.

LOYO Validation

LOYO validation was used to test temporal generalization

Overall $R^2 = 0.692$

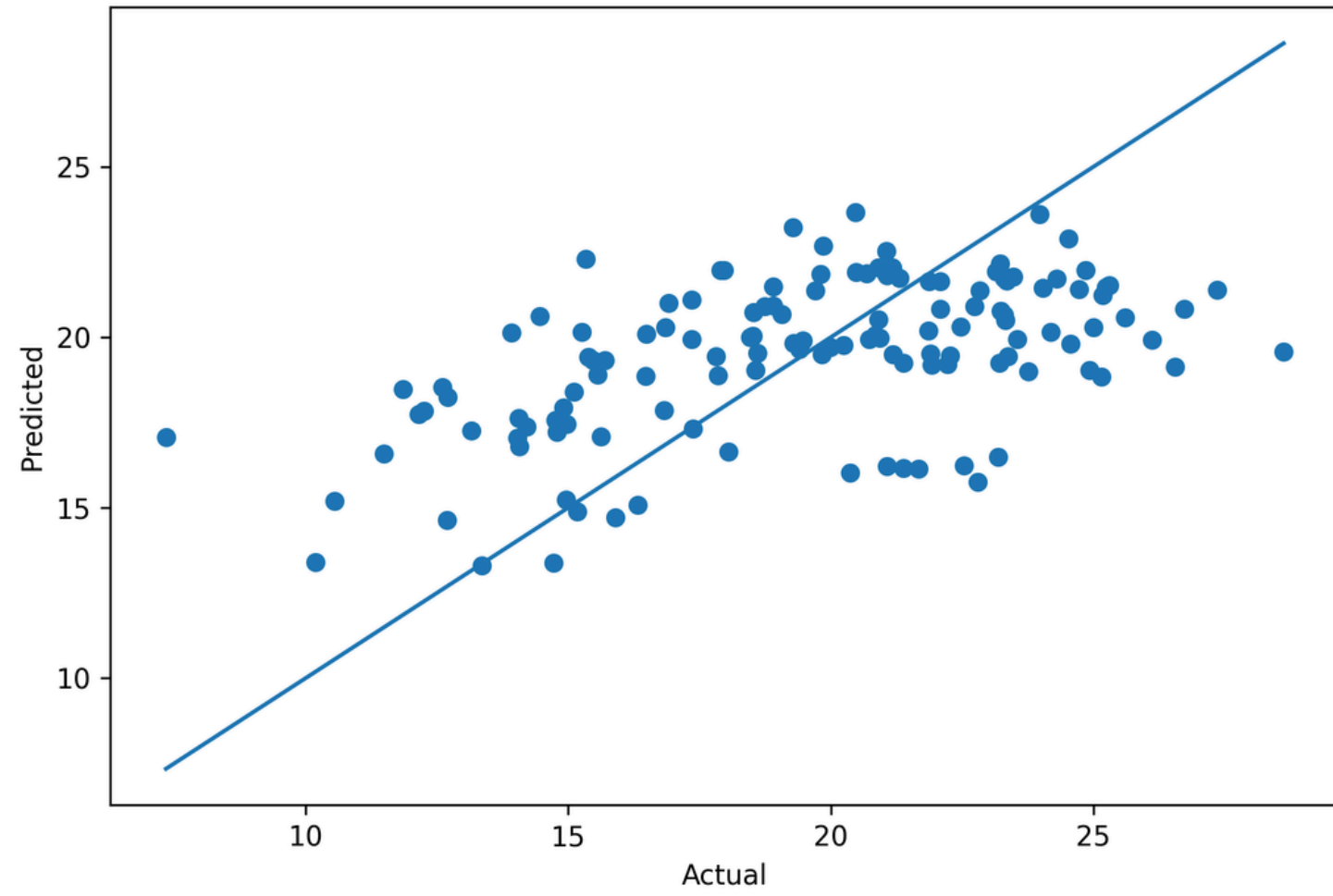
Process:

One year is removed → Model trains on remaining years → Model predicts excluded year

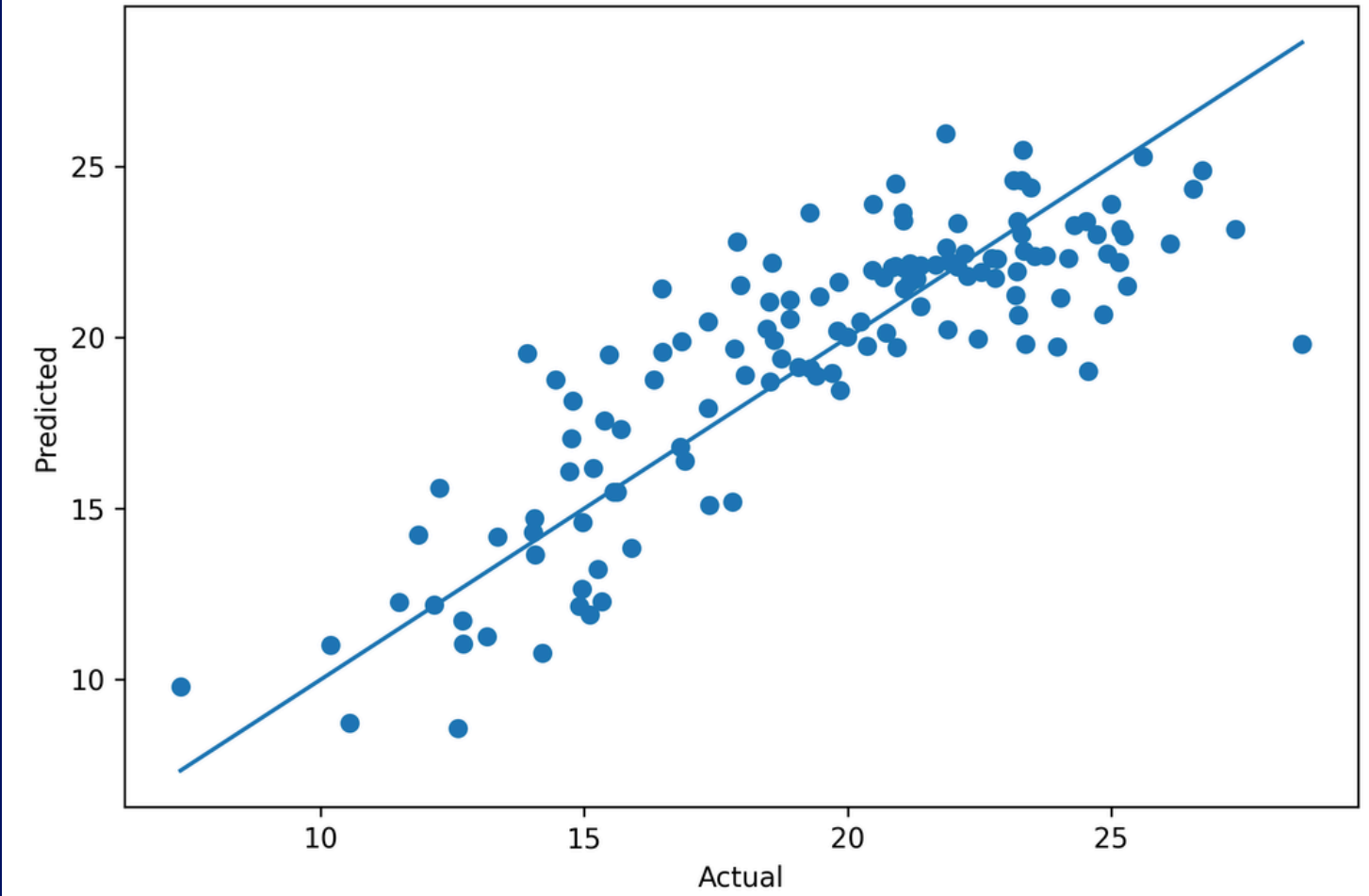
The LOYO validation showed that the model had acceptable temporal generalization and was not only fitting the training data.



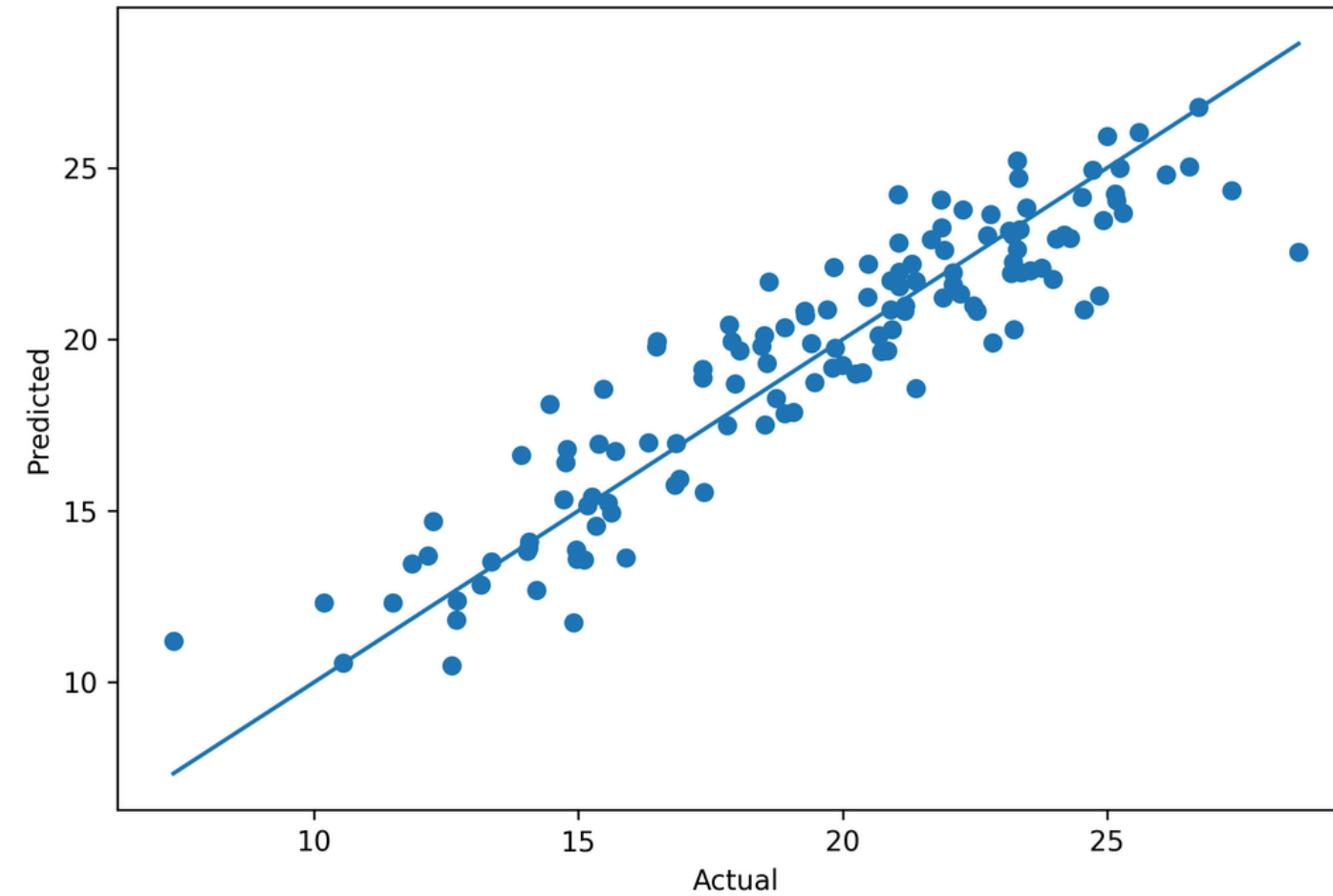
Actual vs Predicted - Pooled OLS



Actual vs Predicted - Leave-One-Year-Out



Actual vs Predicted - Fixed Effects





VIF ANALYSIS

Variance Inflation Factor was used to check whether independent variables were highly correlated.

Interpretation:

- VIF 1–5: Acceptable
- VIF 5–10: Moderate
- VIF > 10: High multicollinearity

The selected regression variables were:

- Establishments per 1,000 people aged 15+
- Population Density
- Schools per 1,000 Population

The selected regression variables showed acceptable VIF levels, indicating no severe multicollinearity problem.

$$VIF_i = \frac{1}{1 - R_i^2}$$





CLUSTERING ANALYSIS & PCA

VISUALIZATION

Clustering was used to group Jordanian governorates based on socioeconomic similarity, while PCA was applied to reduce the data dimensions and visualize the clusters clearly.

1. Clustering Features

The clustering model used several indicators:

Unemployment Rate

Establishments per 1,000 people aged 15+

Population Density

Students per Class

Schools per 1,000 Population

Student Ratio 5–19

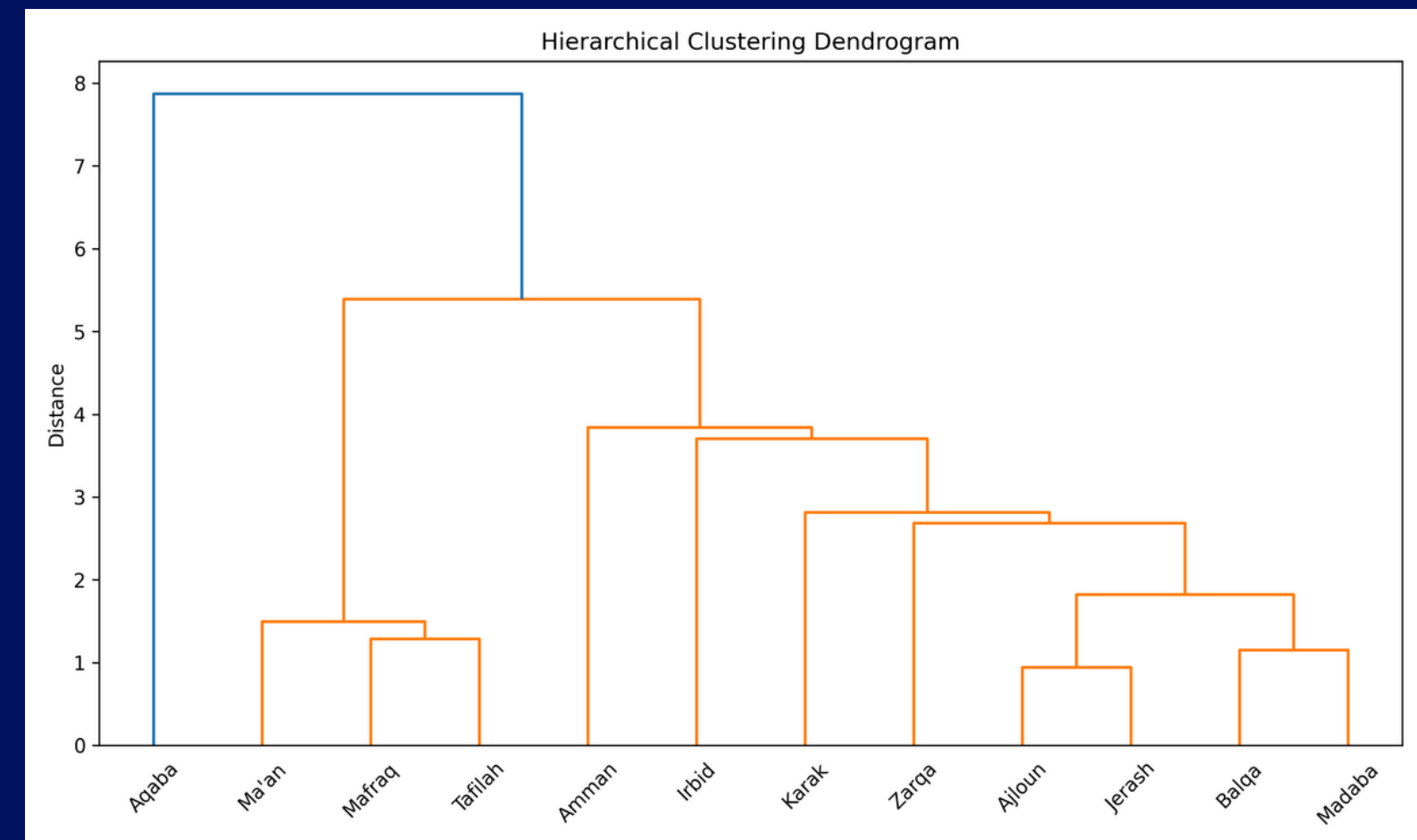


2. Methods Used

Two clustering techniques were applied:

K-Means Clustering: grouped governorates into clusters based on similarity.

Hierarchical Clustering: showed the similarity structure between governorates using a dendrogram.





CLUSTERING ANALYSIS & PCA VISUALIZATION

3. PCA Role

PCA was used to transform multiple variables into two principal components:

$$PC1 = a_1X_1 + a_2X_2 + a_3X_3 + \dots + a_nX_n$$

This made it possible to visualize governorates in a 2D space while preserving the most important variance in the data.

4. Validation Results

The optimal number of clusters was selected using:

Elbow Method

Silhouette Score

Final Result:

Best number of clusters = 2

Key Insight

The clustering results showed that Jordanian governorates can be grouped into two main socioeconomic patterns. PCA helped visualize these groups clearly, and Hierarchical Clustering confirmed the similarity relationships between governorates.



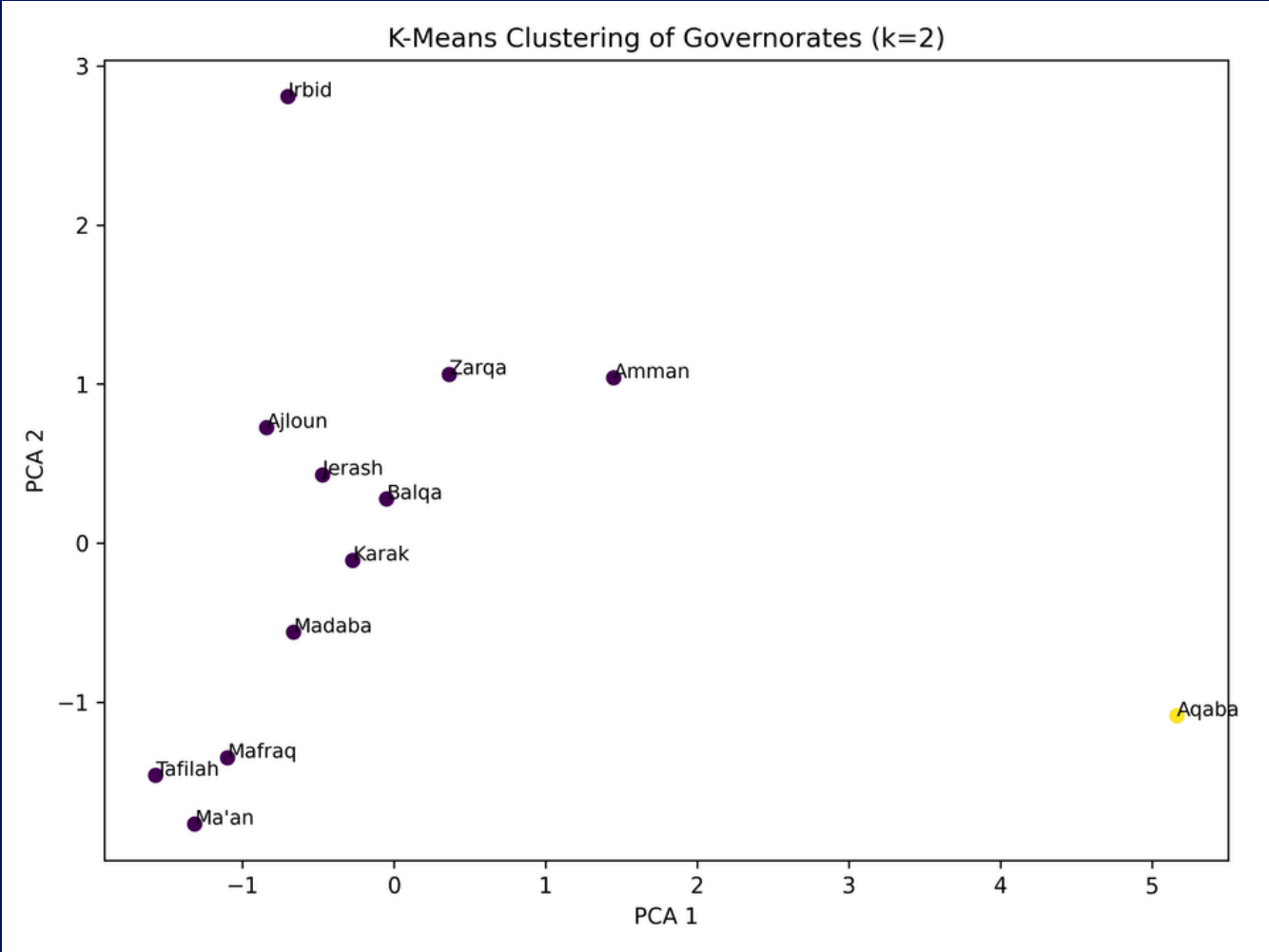
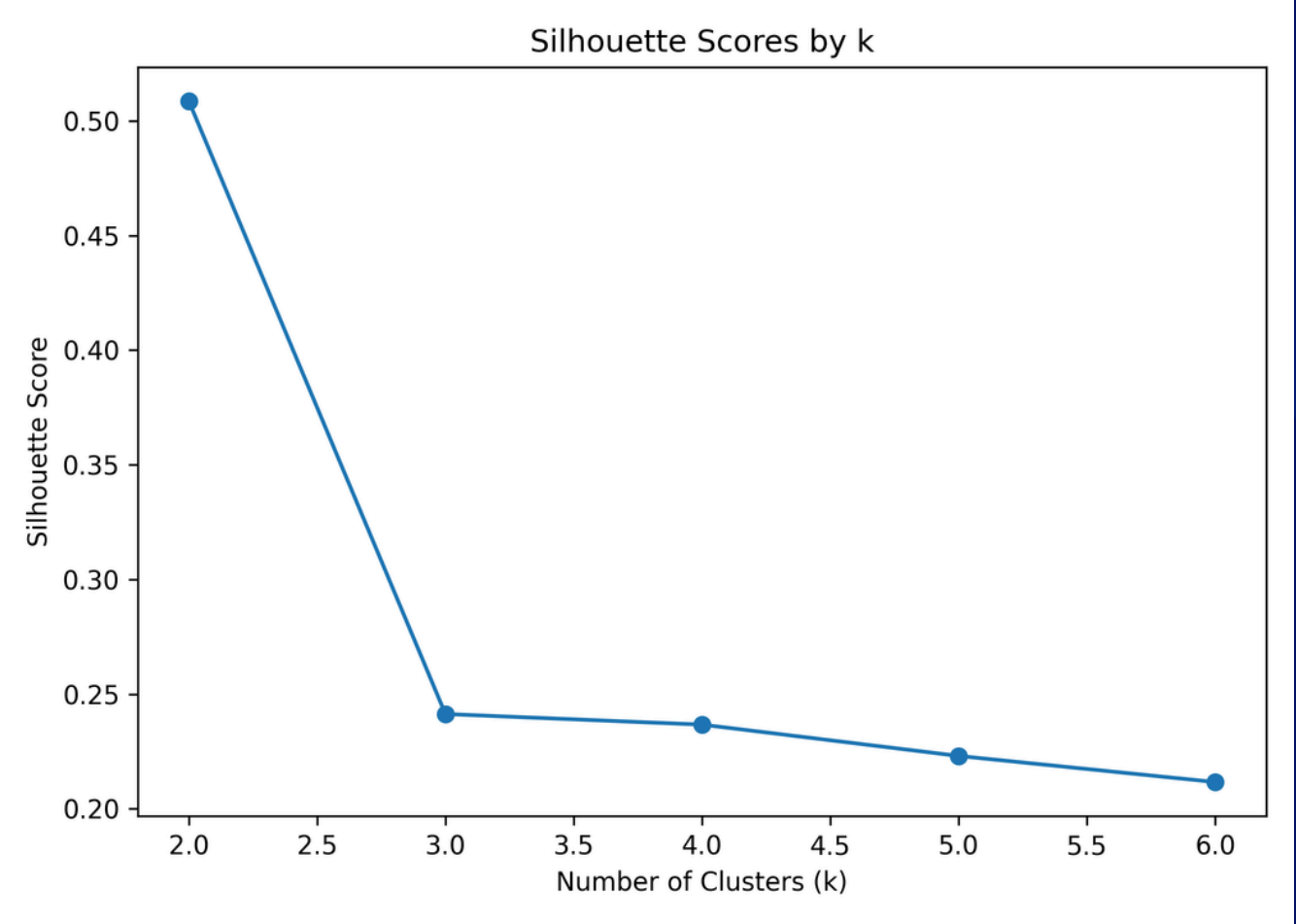
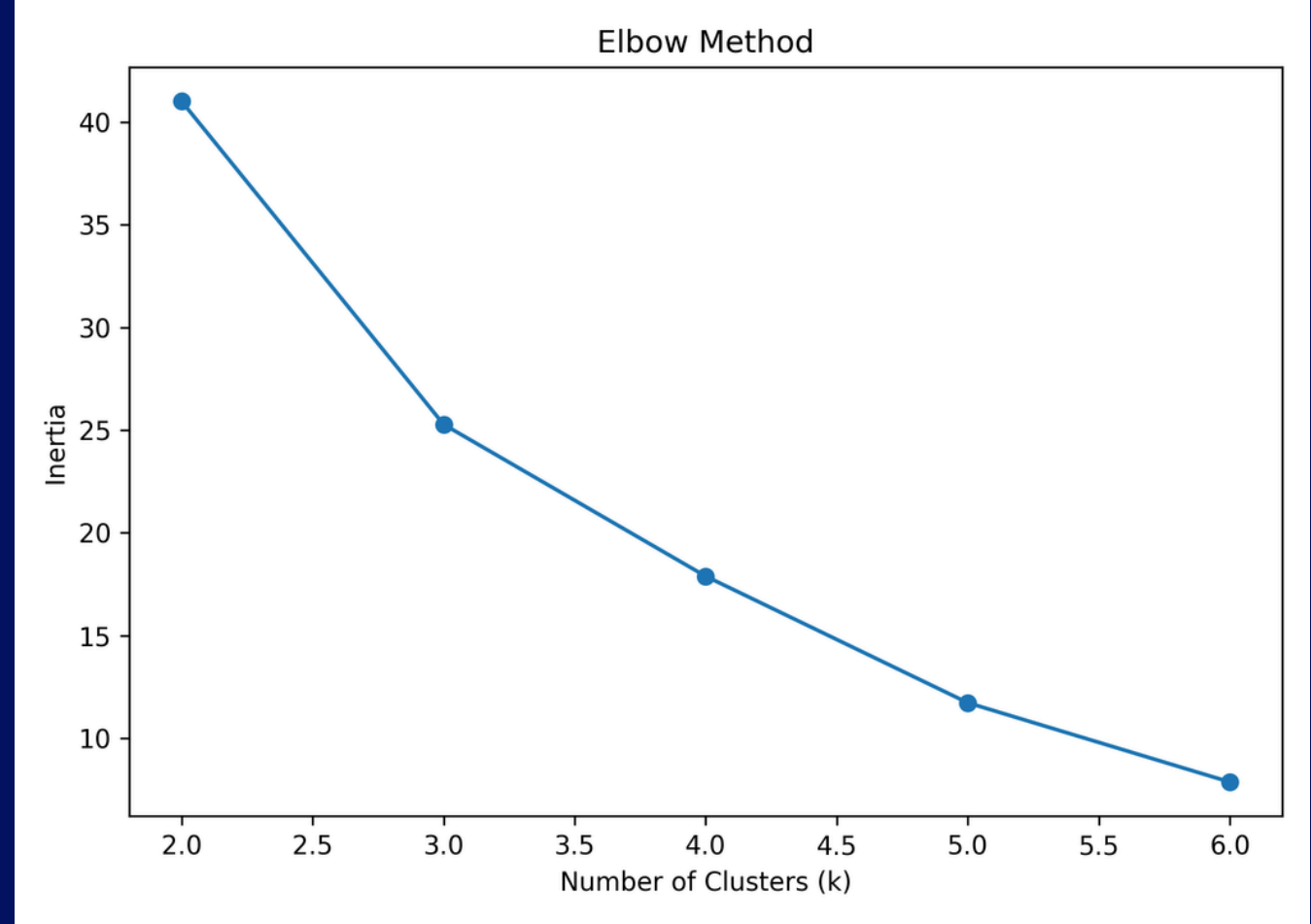


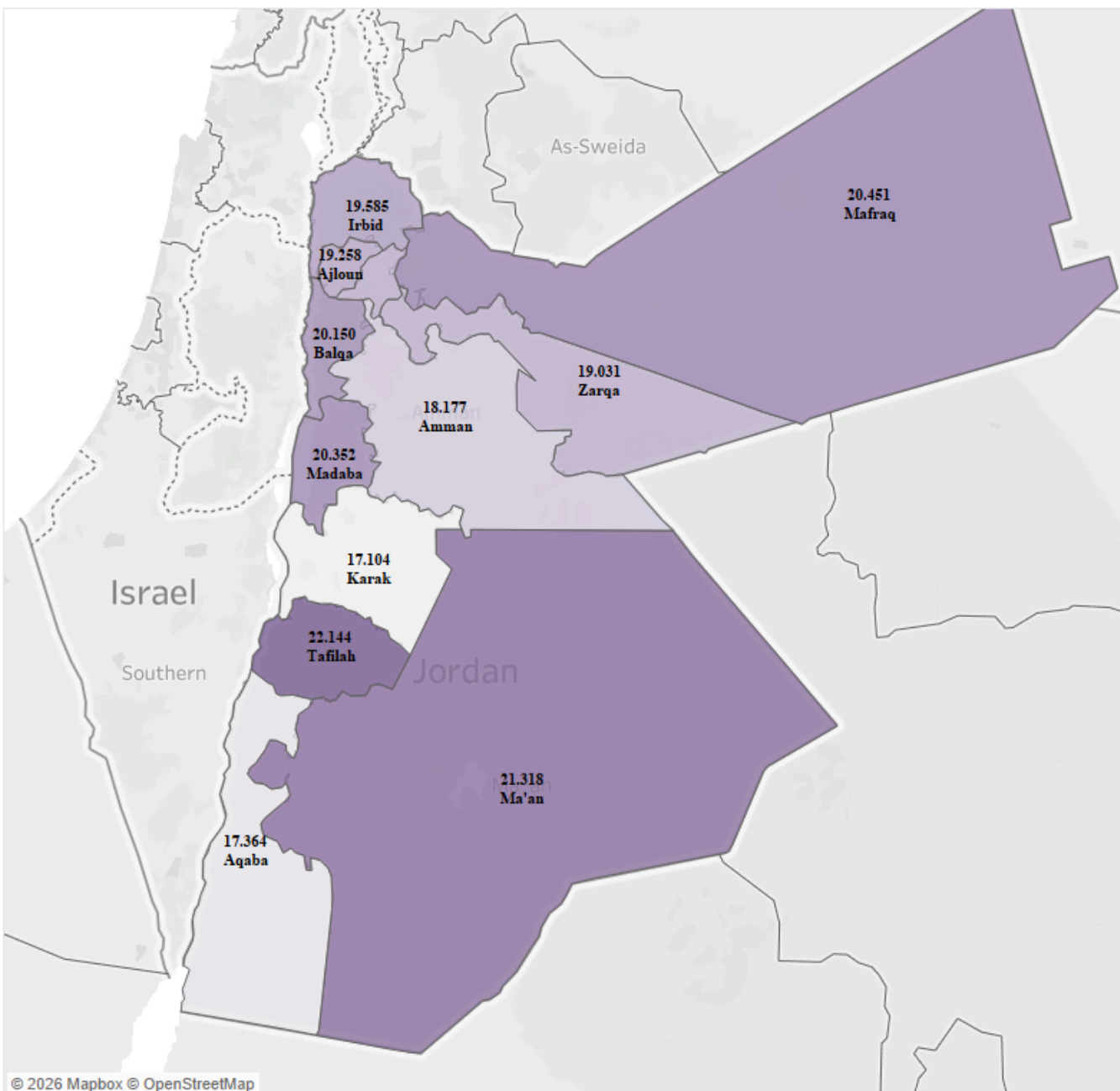


TABLEAU DASHBOARDS

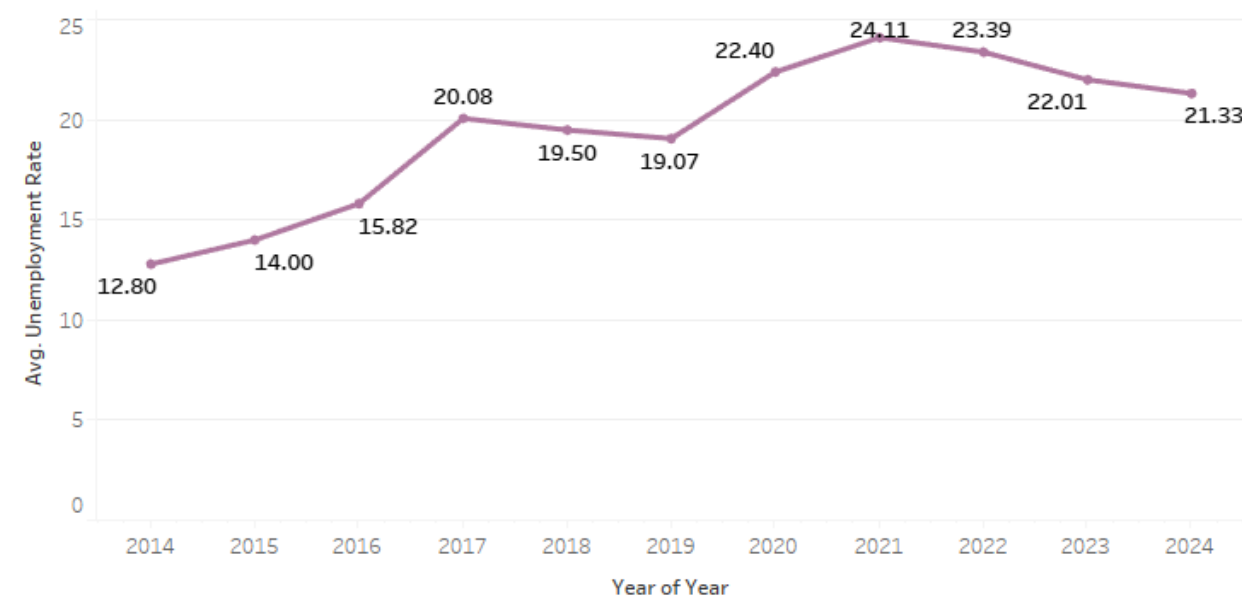
Dashboard 1: Labor Market Overview

Labor Market Overview

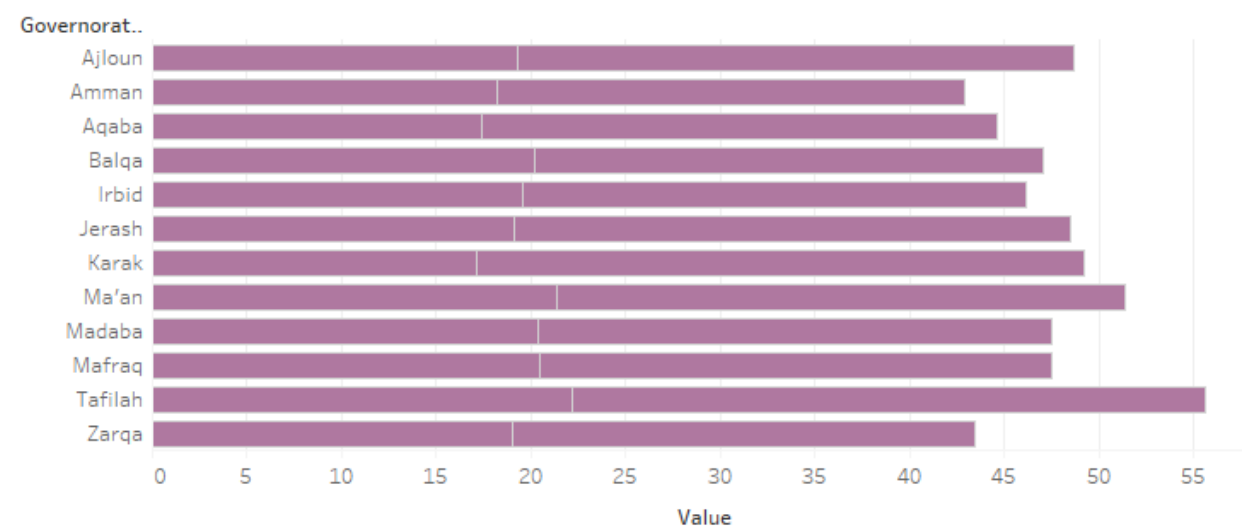
Unemployment Rate by Governorate in Jordan



Unemployment Trend by Year



Employments Rate vs Unemployments Rate



What this dashboard answers

Where is unemployment spatially concentrated?
How did unemployment change from 2014 to 2024?
How do employment and unemployment profiles compare across governorates?

12.8

avg. unemployment in 2014

24.1

peak in 2021

21.3

avg. unemployment in 2024

Main reading

Unemployment rises strongly after 2016 and peaks around 2021.
The map highlights variation across governorates rather than one uniform national pattern.
The employment/unemployment view supports a fast labor-market comparison by governorate.

Dashboard role

This dashboard tells the “what is happening?” part of the story before moving to explanatory factors.





TABLEAU DASHBOARDS

Dashboard 2: Key Determinants

Dashboard structure

Three scatter plots show bivariate visual relationships. Trend lines indicate the direction of each relationship. The coefficient chart summarizes the final model results.

Visual patterns

Establishment density: weak negative visual relationship.
Population density: very weak negative visual relationship.
Schools per 1,000: visible positive bivariate trend in the dashboard.

Model-based interpretation

Strongest coefficient by absolute value: Schools per 1,000 population (-0.1262).
Lightest effect: Population density (-0.0033).
Establishments per 1,000 shows a small positive coefficient ($+0.0500$).

Important note

Scatter plots are descriptive; coefficients provide the model-based evidence.





CONCLUSION & RECOMMENDATIONS

1. What We Found from the Data?

- Unemployment varies clearly across Jordanian governorates.
- Fixed Effects showed that regional differences are important.
- Economic activity, population density, and education indicators are related to unemployment.
- Clustering revealed two main socioeconomic patterns among governorates.

Recommendations

- **Encourage investment in high-unemployment areas.**
- **Improve alignment between education and labor market needs.**
- **Create governorate-specific employment policies.**
- **Use BI dashboards for continuous monitoring and decision-making**





THANK YOU FOR YOUR ATTENTION

ANY QUESTIONS ?

